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RESEARCH ARTICLE

Radio Labeling of Supersub−Division of Path Graphs

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ABSTRACT Motivated by the channel assignment problem, we study the radio labeling of graphs. The radio labeling problem is an important topic in discrete mathematics due to its diverse applications, e.g., frequency assignment in mobile communication systems, signal processing, circuit and sensor network design, etc. A graph labeling problem is an assignment of labels to the vertices or edges (or both) of a graph G that satisfy a mathematical constraint. Radio labeling, a vertex labeling of graphs with non-negative integers, finds an important application in the study of radio channel assignment problems. The maximum label used in a radio labeling is called its span, and the smallest possible span of a radio labeling is called the radio number of a graph. In this area, Chartrand et al. (2005) provided important results by computing the exact values of $rn(G)$ for paths and cycles when k is equal to the diameter for certain cases. In this paper, we determine the radio number $rn(G)$ of G where G is the supersub−division of a path P_n with $n \geq 3$ vertices and a complete bipartite graph $K_{2,\alpha}$ with $\alpha \in \mathbb{N}$.

INDEX TERMS Graph labeling, channel assignment, radio labeling, radio number, path, supersub−division.

I. INTRODUCTION

The introduction of graph labeling in the management of confidential data, the encryption of communications, the allocation of channels, and the promotion of the growth of algorithms led to a golden age of data science and information technologies. In the modern high-tech world, wireless technology is the foundation for every aspect of human communication. By the end of the twentieth century, the vast majority of modern communications were based on wireless network technologies. Graph theory and graph labeling have proven useful in solving a variety of real-world problems, such as radar codes, coding to control missiles, crystallographic analysis, computing signals, determining the structure of molecules, and channel assignment in a wireless network.

Radio frequency assignment problems were introduced by Hale [2] in 1980. This idea was later applied to FM radio stations by Chartrand et al. [3] in 2001. Care was taken in frequency allocation to minimize interference between channels. It is important to assign different frequencies to geographically closed radio stations to avoid interference. By converting a channel assignment into a graph label, the interference graph is used to solve the channel assignment problem.

The interference diagram shows very close transmitters as adjacent vertices and close transmitters as vertices two distances apart. The author [4] suggested that two transmitters that have small interference should receive different channels and two transmitters that have large interference should receive channels that are at least two away from each other.

Frequency assignment motivated radio k-labeling [5]. Suppose that k is a positive number. A radio $k - labeling$ for a graph G is a function, $\varpi : V(G) \rightarrow \{0, 1, 2, 3, \ldots, \}$, such that μ', and μ'' satisfy the following condition: $|\varpi(\mu') - \varpi(\mu'')| \geq k + 1 - d(\mu', \mu'')$, which gives the distance between μ', and μ''. A function ϖ has the $span(\varpi) = max\{\varpi(\mu') - \varpi(\mu'') : \mu', \mu'' \in V(G)\}$, which is denoted by $span(\varpi)$. The minimal span over all k-labels of a graph G is called the ψ_k-number and is denoted by $\psi(G)$.

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For the special case when $k = 1$, the $1 - labeling$ is indeed the conventional vertex coloring and we have $\psi_1 = \chi(G) - 1$, where $\chi(G)$ is the chromatic number of G. Another special case is when $k = 2$, the $2 - labeling$ is the same as the distance two labeling (or $L(2, 1)$-labeling), which has been studied extensively in recent years [6], [7], [8], [9]. ψ_2-number is known as the λ-number of G. When $k = diam(G)$, a k-labeling is called an radio labeling.

Definition 1: [3] A *radio labeling* of a graph G is a mapping $\varpi : V(G) \to \{0, 1, 2, 3, \ldots, \}$ such that for every pair of distinct vertices μ', μ'' of G, $|\varpi(\mu') - \varpi(\mu'')| \geq diam(G) + 1 - d(\mu', \mu'')$. The *radio number* of G is defined as $rn(G) := min_{\varpi} span(\varpi)$ with minimum taken over all radio labelings ϖ of G. A radio labeling ϖ of G is *optimal* if $span(\varpi) = rn(G)$. If $diam(G) = 2$ then $rn(G) = \lambda(G)$.

We chose to study radio labeling of supersub−divided paths because it has practical implications for optimizing communication networks. By understanding how radio labeling schemes can be applied to these graphs, we can improve the efficiency and reliability of data transmission in various real-world scenarios, such as computer networks and transportation systems.

II. RELATED WORK

This section provides a comprehensive overview of four different types of literature. In the field of communications, there are numerous concepts that play a critical role in ensuring efficient and organized radio communications. These concepts include CAP, radio labeling, radio number, and frequency assignment. Authors [10] examined a comprehensive overview of graph labeling and the authors [11] discussed an assignment of radio numbers to triangular and rhombic honeycomb networks. In addition, the study of the radio and antipodal number for many different types of graphs has been the subject of intellectual research by several authors ([12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]).

The study of radio labeling has been the subject of research by several scholars over the past decade, including [23], [24], [25], [26], and [27]. In addition, a previous study determined the radio number for paths and cycles of [25], while the study of the square was performed in the context of paths of [24], and the study of the square was treated in the context of a cycle of [28]. The research conducted in [29] focused on the study results of the radio number for the generalized prism graph. Moreover, in [30] the properties of the generalized gear graph were discussed and the lower bound of its radio number was established. The authors [26] carried out the research on the radio characteristic of graphs associated with cycles.

For $n \geq 6$, [31] has determined the associated radio number, which is the fourth power of P_n, except in cases where n is congruent to 1 modulo 8. The radio number is studied by [32], which focuses on the middle graph with respect to the path $M(P_n)$. Reference [33] recently analyzed the radio number of tree based line and block graphs. Reference [34] conducted a study to discover the bounds of banana graphs. In particular, the value of the bounds for the $\lambda_{3,2,1}$ is studied. Also, the study has determined the maximum limit of the radio number and the average radio number for the (HC_n) and honeycomb torus $(HCT(r))$ honeycomb networks studied by [35]. For more information on the radio numbers and radio antipodal numbers, see ([36], [37], [38], [39], [40], [41]).

The evolution of wireless network infrastructure technologies has contributed significantly to the rapid expansion of mobile interaction and wireless communications programs that adapt to an ever-growing user base. The development of communication overlap and interference can result from issues related to the allocation of spectrum for transmitters and receivers. An efficient allocation algorithm must be developed to effectively manage network constraints and mitigate sporadic communications [42]. In addition, the main objective is to manage the constraints associated with wireless communications, especially in high-density areas where multiple wireless networks coexist ([43], [44]).

Many authors have found the radio labeling and radio labeling of path graphs, the middle graph of the path, a strong product of the path graph, the cross product of the path, and the corona product of the path, etc. The concept of subdivision in the field of graph theory plays a crucial role in facilitating the computation of aspects for complex graphs by relating them to simpler graphs. Subdivision graphs are used to derive various chemical and mathematical properties of complex graphs from simpler graphs. Extensive research has been done by [45] and [46] on these graphs, which facilitate the study of the chemical and physical properties of the object represented by the graph.

This study aims to fill the research gaps by determining the radio number for supersub−division of paths. Furthermore, this study also considers the center of a graph and the total level function to obtain the radio number. In addition, we provide a general discussion on the wireless communication network for the proposed solution, which also improves the clarity of the work.

III. PRELIMINARIES

For basic definitions and concepts of graph theory, see [47]. A graph G is a tuple (V, E), V is a non-empty set of vertices, and $E \subseteq V$. The elements of E are called *edges*. Let $d_G u$ be the *degree* of a vertex u in G (i.e., the number of edges incident with u). Let $P_n = v_1 v_2 v_3 \ldots v_n$ be a *path* of order n. The length of a shortest (a, b)- path in G is denoted by $d_G(a, b)$. The value $e_G(a) = max\{d_G(a, b) | b \in V\}$ is called *eccentricity* of a. The diameter of G, denoted by $diam(G)$ or simply by d, is $max\{d(u, v) : u, v \in V(G)\}$. The *center* of a graph G, denoted by $C(G)$, is the subgraph induced by the vertices with minimum eccentricity.

In 2001, Sethuraman and Selvaraju [48] introduced a graph operation called *supersub-division* of a graph. It is denoted $SSD(G)$ if $SSD(G)$ is obtained from G by replacing each edge xy of G by a complete bipartite graph $K_{2,\alpha}$ by in such a way that the end vertices x, y of each edge are merged with the two vertices of 2-vertices part of $K_{2,\alpha}$ after removing the edge

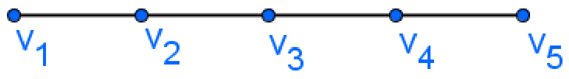

FIGURE 1. Path P_5.

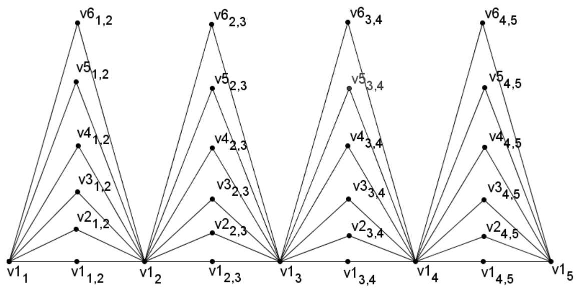

FIGURE 2. The path P_5 and its super subdivision.

TABLE 1. Table of notations.

Parameters	Description
$SSD(G)$	Supersub–division of graphs
V_i^1	Original vertex in G
$V_{i,i+1}^j$	Vertices in $SSD(G)$
$C(SSD(G))$	Center of the $SSD(G)$ graph
$\varepsilon(v)$	Eccentricity of a vertex v
v_C	Vertices in the center of graph $SSD(G)$
v_L	Be the vertices on the left side of the center
v_R	Be the vertices on the right side of the center
$\ell(u)$	Level function
$\ell(G)$	Total level function.
$rn(G)$	Radio number.

xy from G. Thus the number of vertices is $n + \alpha q$. Thus, $|V(SS(G))| = n+\alpha q$ and $|E(SS(G))| = 2\alpha q$. In the following figures 1 and 2, the graph $G = P_5$ and its super subdivision by $K_{2,6}$ are shown.

Let the vertex set of supersub–division of a path be $V = \{v_i^1, v_{i,i+1}^j : 1 \le i \le n-1, 1 \le j \le \alpha\} \cup \{v_n^1\}$ For example (2), if $n = 5$ and $\alpha = 6$ then the labeling is as follows:

IV. NOTATION

Lemma 1: For $n \ge 1$ and $j \ge 1$,

$$C(SSD(P_n)) = \begin{cases} v^j_{\frac{n}{2},\frac{n}{2}+1} & \text{if } n \text{ is an even.} \\ v_{\frac{n+1}{2}} & \text{if } n \text{ is an odd.} \end{cases}$$

Proof: Let $SSD(P_n)$ be supersub–division of a path P_n and S be a collection of all vertices in $SSD(P_n)$ that have the smallest possible eccentricity. The eccentricity for both the vertex $v_{i,i+1}^j$, $1 \le i \le n-1$, $1 \le j \le \alpha$ and the vertex v_b^1, $1 \le b \le n$ can be expressed as follows:

Case (1) (If n an odd number:)

$$\varepsilon(v_b^1) = (n-1) + 2\left|\frac{n+1}{2} - b\right|, \quad 1 \le b \le n.$$

$$\varepsilon(v_{i,i+1}^j) = (n-1) + \left|\frac{n-1}{2} - i\right| + \left|\frac{n+1}{2} - i\right|, \quad 1 \le i \le n-1, \text{ and } 1 \le j \le \alpha.$$

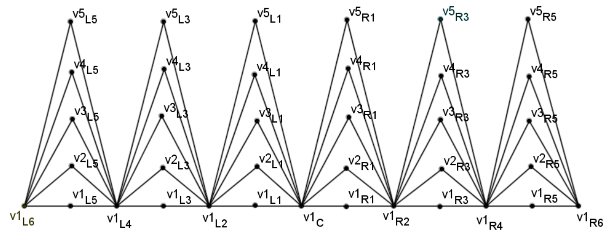

FIGURE 3. $SSD(P_7)$ where $\alpha = 5$.

One can see that S contains only $v_{\frac{n+1}{2}}$. This is because according to the above function, $\varepsilon(v_{\frac{n+1}{2}})$ is smaller than the eccentricity of the other vertices in $SSD(P_n)$. There is no doubt that case (1) is $C(SSD(P_n)) = v_{\frac{n+1}{2}}$.

Case (2) (If n an even number:)

$$\varepsilon(v_b^1) = (n-1) + \left\{\left|\frac{n}{2} - b\right| + \left|\frac{n+2}{2} - b\right|\right\}, \quad 1 \le b \le n.$$

$$\varepsilon(v_{i,i+1}^j) = (n-1) + \left\{2\left|\frac{n}{2} - i\right|\right\}, \quad 1 \le i \le n-1, 1 \le j \le \alpha.$$

One can see that S contains $v^j_{\frac{n}{2},\frac{n}{2}+1}$ where $1 \le j \le \alpha$. This is because according to the above function $\varepsilon(v^j_{\frac{n}{2},\frac{n}{2}+1})$ is smaller than the eccentricity of the remaining vertices in $SSD(P_n)$. There is no doubt that the case (2) is $C(SSD(P_n)) = v^j_{\frac{n}{2},\frac{n}{2}+1}$.

Let us now consider the new designation of the vertex of $SSD(P_n)$ with respect to its center. In the further course of this paper, we will use the following labels for the designations.

If n is an odd number, then let $n = 2k+1$

$$v_b^1 = \begin{cases} v_C^1 = v_b^1 & b = k+1, \\ v_{L2(k+1-b)} & 1 \le b \le k, \\ v_{R2(b-k-1)} & k+2 \le b \le 2k+1. \end{cases}$$

$$v_{i,i+1}^j = \begin{cases} v^j_{L2(k+1-i)-1} & 1 \le i \le k, 1 \le j \le \alpha, \\ v^j_{R2((i+1)-(k+1))-1} & k+1 \le i \le 2k, 1 \le j \le \alpha. \end{cases}$$

If n is an even number, then let $n = 2k$

$$v_b^1 = \begin{cases} v_{L2(k-b)+1} & 1 \le b \le k, \\ v_{R2(b-k)-1} & k+1 \le b \le 2k. \end{cases}$$

$$v_{i,i+1}^j = \begin{cases} v_C^j = v^j_{C(k,k+1)} & 1 \le j \le \alpha, k = \dfrac{n}{2}, \\ v^j_{L2(k-i)} & 1 \le i \le k, 1 \le j \le \alpha, \\ v^j_{R2(i-k)} & k+1 \le i \le 2k-1, 1 \le j \le \alpha. \end{cases}$$

Illustration: In Figures (3) and (4), the supersub–division $SS(P_7^5)$and $SS(P_8^5)$.

One defines the level function to $V(SSD(P_n))$ as the set of whole number $\mathbb{W}$ by $\ell(\mu) = min\{d(\mu, \mu'); \mu' \in V(C(SSD(P_n)))\}$ for each $\mu \in V(SSD(P_n)) - V(C(SSD(P_n)))$. In the graph $SSD(P_n)$, the maximum level is $n-1$. In the graph G, $\ell(G)$ represents the total level, which is defined as follows: $\ell(G) = \sum_{\mu \in V(G)} \ell(\mu)$.

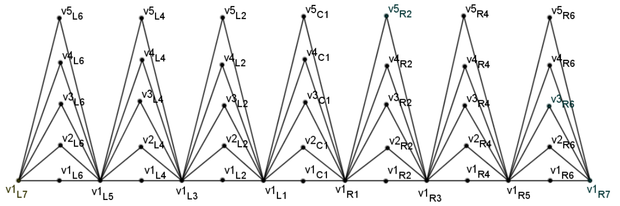

FIGURE 4. ***SSD*(P_8) where $\alpha = 5$.**

V. MAIN RESULT

Chartrand, Erwin and Zhang [1] in their lecture on paths, also called P_n, established the upper bounds of $rn(P_n)$ for paths on n vertices.

Theorem 1: [1] *For* $n \geq 1$,

$$rn(P_n)) \leq \begin{cases} 2k^2 + k & if\ n = 2k + 1; \\ 2(k^2 - k) + 1 & if\ n = 2k. \end{cases}$$

Moreover, the bounds is sharp where $n \geq 5$.

Liu and Zhu [25] shown the radio number of paths in the following result.

Theorem 2: [25] *For* $n \geq 3$,

$$rn(P_n)) = \begin{cases} 2k(k-1) + 1 & if\ n = 2k; \\ 2k^2 + 2 & if\ n = 2k + 1. \end{cases}$$

In this paper, we found radio labeling of supersub–division of paths of length n (P_n), where n is positive integer determined and also we found center of supersub–division of paths.

Remark 1: (a) $|V(SSD(P_n))| = n + \alpha q$

(b) $d(\mu, \mu') \leq \ell(\mu) + \ell(\mu')$

Lemma 2: *For* $n \geq 1$,

$$\ell(SSD(P_n)) = \begin{cases} \dfrac{n^2(\alpha+1) + \alpha(1-2n) - 1}{2} & \text{if } n \text{ is an odd.} \\ \dfrac{\alpha n(n-2) + n^2}{2} & \text{if } n \text{ is an even.} \end{cases}$$

Proof: If n is an odd number, the vertex $v_{\frac{n+1}{2}}$ is a enter of $SSD(P_n)$. i.e., $C(SSD(P_n)) = v_{\frac{n+1}{2}}$ if n is an odd.

$$\begin{aligned} \sum_{\mu \in V(SSD(P_n))} \ell(\mu) &= 2\left(\alpha \sum_{k=1}^{\frac{n-1}{2}} 2k - 1 + \sum_{k=1}^{\frac{n-1}{2}} 2k\right) \\ &= 2\left(\begin{array}{c} \alpha\left(2\left(\frac{\frac{n-1}{2}(\frac{n-1}{2}+1)}{2}\right) - \frac{n-1}{2}\right) \\ +2\left(\frac{(\frac{n-1}{2})(\frac{n-1}{2}+1)}{2}\right) \end{array}\right) \\ &= 2\left((\alpha+1)\left(\frac{(n-1)(n+1)}{4}\right) - \frac{\alpha(n-1)}{2}\right) \\ &= 2\left(\frac{\alpha n^2 - \alpha + n^2 - 1 - 2\alpha n + 2\alpha}{4}\right) \\ &= \frac{n^2(\alpha+1) + \alpha(1-2n) - 1}{2}. \end{aligned}$$

If n is an odd number, the vertex $v^j_{\frac{n}{2}, \frac{n}{2}+1}$ is a enter of $SSD(P_n)$. i.e., $C(SSD(P_n)) = v^j_{\frac{n}{2}, \frac{n}{2}+1}$ if n is an even.

$$\begin{aligned} \sum_{\mu \in V(SSD(P_n))} \ell(\mu) &= 2\left(\sum_{i=1,3,5,\ldots,}^{n-1} i + \alpha \sum_{i=2,4,6,\ldots,}^{n-2} i\right) \\ &= 2\left(\alpha \sum_{k=1}^{\frac{n-2}{2}} 2k + \sum_{k=1}^{\frac{n}{2}} 2k - 1\right) \\ &= 2\left(\begin{array}{c} 2\alpha\left(\frac{\frac{n-2}{2}\left(\frac{n-2}{2}+1\right)}{2}\right) \\ +2\left(\frac{\frac{n}{2}\left(\frac{n}{2}+1\right)}{2}\right) - \frac{n}{2} \end{array}\right) \\ &= 2\left(\frac{\alpha n(n-2) + (n+2)n - 2n}{4}\right) \\ &= \frac{\alpha n(n-2) + n^2}{2}. \end{aligned}$$

Theorem 3: Let ϖ *be an assignment of distinct non–negative integer to* $V(SSD(P_n))$ *and* $\mu_1, \mu_2, \mu_3, \ldots, \mu_\delta$ *be the ordering of* $V(SSD(P_n))$ *such that* $\varpi(\mu_i) < \varpi(\mu_{i+1})$ *defined by* $\varpi(\mu_1) = 0$ *and* $\varpi(\mu_{i+1}) = \varpi(\mu_i) + d + 1 - d(\mu_i, \mu_{i+1})$. *If* $d(\mu_i, \mu_{i+1}) \leq n$ *for any* $1 \leq i \leq \delta - 1$. *Then* ϖ *is radio labeling.*

Proof: Let ϖ be an assignment of distinct non-negative integers to $V(G)$ such that $\varpi(\mu_1) = 0$, $\varpi(\mu_{i+1}) = \varpi(\mu_i) + d + 1 - d(\mu_i, \mu_{i+1})$, for any $1 \leq i \leq \delta - 1$ holds.

We intend to provide proof for ϖ is radio labeling. i.e., for any $i \neq j$, for each $1 \leq i \leq \delta - 1$, then consider $\varpi_i = \varpi(\mu_{i+1}) - \varpi(\mu_i)$. We take $i > j$, then

$$\begin{aligned} \varpi(\mu_i) - \varpi(\mu_j) &= \varpi_j + \varpi_{j+1} + \varpi_{j+2} +, \ldots + \varpi_{i-2} + \varpi_{i-1} \\ &= (i-j)(d+1) - d(\mu_j, \mu_{j+1}) - \\ &\quad d(\mu_{j+1}, \mu_{j+2}) -, \ldots, -d(\mu_{i-1}, \mu_i) \\ &\geq (i-j)(d+1) - (i-j)(n-1) \\ &\geq (i-j)(d+1-(n-1)) \\ \varpi(\mu_i) - \varpi(\mu_j) &\geq d + 1 - d(\mu_i, \mu_j) \end{aligned}$$

Since $i - j \neq 0$. Hence, ϖ is a radio labeling.

Theorem 4: For $n_1 \geq 1$ *and* $n = n_1 + 2$, $rn(K_{2,n_1}) = n$.

Proof: Let G be complete bipartite graph with vertex partition V_1 and V_2 with n_1 and n_2 elements. One can seen that diameter of G is 2. Now label all the vertex from any one of the partition consecutive numbers from 0 to one lesser of number of elements in respective partition and start the labeling to the other partition consecutive numbers from greater one of previous partition to order of G. Hence, this satisfies radio labeling condition and providing the minimality by diameter of G is 2. From the above theorem, we have following.

Corollary 1: For $\alpha \geq 1$, $rn(SSD(P_2)) = 2 + \alpha$.

Proof: Since $SSD(P_n)$ is isomorphic to $K_{2,n}$, the proof follows from the theorem (4) and illustration shown in (5).

Theorem 5: Let $n \geq 4$ *be an even and* $\alpha \geq 1$. *Then,* $rn(SSD(P_n)) \geq n^2 - 3n - 2\alpha - \alpha q - n^2\alpha + 2\alpha n + 2\alpha n q + 5$.

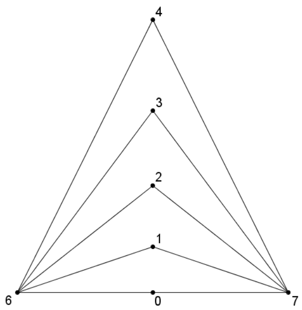

FIGURE 5. $SSD(P_2) = 7$ where $\alpha = 5$.

Proof. Let $n \geq 4$ be an even and $\alpha \geq 1$. Let ϖ be an assignment of unique non-negative integers to $V(SSD(P_n))$ and $\mu_1, \mu_2, \mu_3, \ldots, \mu_\delta$ be the order of $V(SSD(P_n))$ such that $\varpi(\mu_i) < \varpi(\mu_{i+1})$ defined by $\varpi(\mu_1) = 0$ and $\varpi(\mu_{i+1}) = \varpi(\mu_i) + d + 1 - d(\mu_i, \mu_{i+1})$. By theorem 3 ϖ is the radio labeling of $SSD(P_n)$, then

$$\sum_{i=1}^{\delta-1} \varpi(\mu_{i+1}) - \varpi(\mu_i) \geq \sum_{i=1}^{\delta-1}(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1})$$

$$\varpi(\mu_\delta) - \varpi(\mu_1) \geq (\delta-1)(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1})$$

$$rn(SSD(P_n)) \geq (\delta-1)(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1})$$

By remark 1,

$$\begin{aligned}
d(\mu_i, \mu_{i+1}) &\leq \ell(\mu_i) + \ell(\mu_{i+1}) \\
\sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1}) &\leq \sum_{i=1}^{\delta-1}[\ell(\mu_i) + \ell(\mu_{i+1})] \\
&= [\ell(\mu_1) + \ell(\mu_2)] + [\ell(\mu_2) + \ell(\mu_3)] \\
&\quad +, \ldots, +[\ell(\mu_{\delta-1}) + \ell(\mu_\delta)] \\
&= [\ell(\mu_1) + \ell(\mu_2) + \ell(\mu_3)+, \ldots, \ell(\mu_{\delta-1})] \\
&\quad + [\ell(\mu_2) + \ell(\mu_3)+, \ldots, +\ell(\mu_\delta)] \\
&= \sum_{\mu \in V(G)} \ell(\mu) - \ell(\mu_\delta) \\
&\quad + \sum_{\mu \in V(G)} \ell(\mu) - \ell(\mu_1) \\
&= 2 \sum_{\mu \in V(G)} \ell(\mu) - \ell(\mu_1) - \ell(\mu_\delta) \\
&= 2\ell(SSD(P_n))
\end{aligned}$$

Hence, by Lemma 2. (choosing $\mu_1 \in V(C(SSD(P_n)))$and $\mu_\delta \in V(C(SSD(P_n))))$, $\ell(\mu_1) = 0$, $\ell(\mu_\delta) = 0$.

$$\begin{aligned}
rn(SSD(P_n)) &\geq (\delta-1)(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1}) \\
&= (\delta-1)(d+1) - (2\ell(SSD(P_n)) + 2(m-2)) \\
&= (\delta-1)(d+1) \\
&\quad - \left(2\left(\frac{\alpha n(n-2) + n^2}{2}\right) + 2(m-2)\right) \\
&= (n + \alpha q - 1)(2n - 2 + 1) \\
&\quad - 2\left(\frac{\alpha n(n-2) + n^2}{2}\right) - 2(m-2) \\
&= n^2 - 3n - 2\alpha - \alpha q - n^2\alpha + 2\alpha n + 2\alpha n q + 5.
\end{aligned}$$

Theorem 6: Let $n \geq 3$ be an odd and $\alpha \geq 1$. Then $rn(SSD(P_n)) \geq n^2 - 3n - \alpha q - \alpha n^2 + 2\alpha n q - \alpha + 2\alpha n + 4$.

Proof: Let $n \geq 3$ be an odd and $\alpha \geq 1$. Let ϖ be an assignment of unique distinct non−negative integers to $V(SSD(P_n))$ and $\mu_1, \mu_2, \mu_3, \ldots, \mu_\delta$ be the order of $V(SSD(P_n))$ such that $\varpi(\mu_i) < \varpi(\mu_{i+1})$ defined by $\varpi(\mu_1) = 0$ and $\varpi(\mu_{i+1}) = \varpi(\mu_i) + d + 1 - d(\mu_i, \mu_{i+1})$. By theorem 3 ϖ is the radio labeling of $SSD(P_n)$, then

$$\sum_{i=1}^{\delta-1} \varpi(\mu_{i+1}) - \varpi(\mu_i) \geq \sum_{i=1}^{\delta-1}(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1})$$

$$\varpi(\mu_\delta) - \varpi(\mu_1) \geq (\delta-1)(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1})$$

$$rn(SSD(P_n)) \geq (\delta-1)(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1})$$

By remark 1,

$$\begin{aligned}
d(\mu_i, \mu_{i+1}) &\leq \ell(\mu_i) + \ell(\mu_{i+1}) \\
\sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1}) &\leq \sum_{i=1}^{\delta-1}[\ell(\mu_i) + \ell(\mu_{i+1})] \\
&= [\ell(\mu_1) + \ell(\mu_2)] + [\ell(\mu_2) + \ell(\mu_3)]+, \ldots, \\
&\quad + [\ell(\mu_\delta - 1) + \ell(\mu_\delta)] \\
&= [\ell(\mu_1) + \ell(\mu_2) + \ell(\mu_3)+, \ldots, \ell(\mu_\delta - 1)] \\
&\quad + [\ell(\mu_2) + \ell(\mu_3)+, \ldots, +\ell(\mu_\delta)] \\
&= \sum_{\mu \in V(G)} \ell(\mu) - \ell(\mu_\delta) + \sum_{\mu \in V(G)} \ell(\mu) - \ell(\mu_1) \\
&= 2 \sum_{\mu \in V(G)} \ell(\mu) - \ell(\mu_1) - \ell(\mu_\delta) \\
&= 2\ell(SSD(P_n))
\end{aligned}$$

Hence, by Lemma 2. (choosing $\mu_1 \in V(C(SSD(P_n)))$and $\mu_\delta \in N(\mu_1))$, $\ell(\mu_1) = 0$, $\ell(\mu_\delta) = 1$.

$$\begin{aligned}
rn(SSD(P_n)) &\geq (\delta-1)(d+1) - \sum_{i=1}^{\delta-1} d(\mu_i, \mu_{i+1}) \\
&= (\delta-1)(d+1) - (2\ell(SSD(P_n))) \\
&= (\delta-1)(d+1) \\
&\quad - \left(2\left(\frac{n^2(\alpha+1) + \alpha(1-2n) - 1}{2}\right)\right) \\
&= (n + \alpha q - 1)(2n - 2 + 1) \\
&\quad - 2\left(\frac{n^2(\alpha+1) + \alpha(1-2n) - 1}{2}\right) \\
&= n^2 - 3n - \alpha q - \alpha n^2 + 2\alpha n q - \alpha + 2\alpha n + 3.
\end{aligned}$$

Theorem 7: For $\alpha \in \mathbb{N}$, Then

$$rn(SSD(P_n)) \leq \begin{cases} n^2 - 3n - 2\alpha - \alpha q - \alpha n^2 + 2\alpha n \\ +2\alpha nq + 5 & \textit{if } n \textit{ is an even.} \\ n^2 - 3n - \alpha q - \alpha n^2 + 2\alpha nq - \alpha \\ +2\alpha n + 4 & \textit{if } n \textit{ is an odd.} \end{cases}.$$

Proof: To demonstrate the required conclusion, we will examine two distinct cases. *Case (1) (When n is an even number:)*
For $SSD(P_n)$, let
$\varpi : V(SSD(P_n)) \rightarrow \{0, 1, 2, \ldots, (n^2 - 3n - \alpha q - \alpha n^2 + 2\alpha n + 2\alpha nq + 1)\}$ is defined by
$\varpi(\mu_{i+1}) = \varpi(\mu_i) + d + 1 - \ell(\mu_i) - \ell(\mu_{i+1})$ for $i = 1, 2, \ldots, \delta - 2$ and
$\varpi(\mu_\delta) = \varpi(\mu_{\delta-1}) + d - \ell(\mu_{\delta-1}) - \ell(\mu_\delta)$.

We first set $\mu_1 = v^1_{(\frac{n}{2}, \frac{n}{2}+1)}$, $\mu_\delta = v^\alpha_{(\frac{n}{2}, \frac{n}{2}+1)}$. By remark 1. $\delta = n + \alpha q$, $v^{\alpha-j}_{(\frac{n}{2}, \frac{n}{2}+1)} = \mu_{\delta-j}$, for $2 \leq i \leq \delta - 1$, and $1 \leq j \leq \alpha$, set $\mu_h := v^1_i$, where

$$h = \begin{cases} (\frac{n}{2} - i)(2\alpha + 2) + 3, & 1 \leq i \leq \frac{n}{2}, \\ (n - i)(2\alpha + 2) + 2, & \frac{n}{2} + 1 \leq i \leq n. \end{cases}$$

and $\mu_h := v^j_{i,i+1}$, where

$$h = \begin{cases} (2j + 3) + (2\alpha + 2) \\ +(\frac{n}{2} - 1 - i), & 1 \leq i \leq \frac{n}{2} - 1,\ 1 \leq j \leq \alpha. \\ (2\alpha + 2)(n - 1 - i) \\ +(2j + 2), & \frac{n}{2} \leq i \leq n,\ 1 \leq j \leq \alpha. \end{cases}$$

Consequently, in each of the previously mentioned scenarios, a linear order $\mu_1, \mu_2, \mu_3, \ldots, \mu_\delta$ was defined for the vertices of $SSD(P_n)$.

According to the arrangement of the vertices indicated below:

$$v^1_C \xrightarrow{\text{n-1}} v^1_{Rk} \xrightarrow{n} v^1_{L1} \xrightarrow{\text{n-1}} v_{R(k-1)} \xrightarrow{\text{n}} v^1_{L2} \xrightarrow{\text{n}} v^2_{R(k-1)} \xrightarrow{\text{n}} v^2_{L2}$$
$$\xrightarrow{\text{n}}, \ldots, \xrightarrow{\text{n}} v^\alpha_{R(k-1)} \xrightarrow{\text{n}} v^\alpha_{L2} \xrightarrow{\text{n-1}} v^1_{R(k-2)} \xrightarrow{\text{n}} v^1_{L3} \xrightarrow{\text{n}} v^1_{R(k-3)}$$
$$\xrightarrow{\text{n}} v^1_{L4} \xrightarrow{\text{n-1}} v^2_{R(k-3)} \xrightarrow{\text{n}} v^2_{L4} \xrightarrow{\text{n}}, \ldots, \xrightarrow{\text{n}} v^\alpha_{R(k-3)} \xrightarrow{\text{n}} v^\alpha_{L4}, \ldots,$$
$$\xrightarrow{\text{n-1}} v^1_{R3} \xrightarrow{\text{n}} v^1_{L(k-2)} \xrightarrow{\text{n-1}} v^1_{R2} \xrightarrow{\text{n}} v^1_{L(k-1)} \xrightarrow{\text{n}} v^2_{R1} \xrightarrow{\text{n}} v^2_{L(k-2)}$$
$$\xrightarrow{\text{n}}, \ldots, \xrightarrow{\text{n}} v^\alpha_{R1} \xrightarrow{\text{n}} v^\alpha_{L(k-1)} \xrightarrow{\text{n-1}} v^1_{R1} \xrightarrow{\text{n}} v^1_{Lk} \xrightarrow{\text{n-1}} v^2_C$$
$$\xrightarrow{2} v^3_C, \ldots, \xrightarrow{2} v^\alpha_C.$$

Case (2) (When n is an odd number:) For $SSD(P_n)$, let
$\varpi : V(SSD(P_n)) \rightarrow \{0, 1, 2, 3, \ldots, (n^2 - 2n - \alpha q - \alpha n^2 + 2\alpha nq - \alpha + 2\alpha n + 1)\}$ is defined by
$\varpi(\mu_{i+1}) = \varpi(\mu_i) + d - \ell(\mu_i) - \ell(\mu_{i+1})$.

We first set $\mu_1 = v^1_{\frac{n+1}{2}}$, $\mu_2 = v^1_n$, and $\mu_3 = v^1_1$ and for $3 \leq i \leq \delta - 1$, set $\mu_h := v^1_i$, where

$$h = \begin{cases} i(2\alpha + 2) - 2\alpha + 1, & 1 \leq i \leq \frac{n-1}{2}, \\ 2i - (n+1)(m+1) + 2, & \frac{n+3}{2} < i \leq n - 1, \end{cases}$$

and $\mu_h := v^\alpha_{i,i+1}$, where

$$h = \begin{cases} (i-1)(2\alpha + 2) + 2j + 3, & 1 \leq i \leq \frac{n-1}{2}, \\ & 1 \leq j \leq \alpha. \\ (2i - n - 1)(\alpha + 1) + 2j + 2, & \frac{n+1}{2} < i \leq n - 1, \\ & 1 \leq j \leq \alpha. \end{cases}$$

Consequently, in each of the previously mentioned scenarios, a linear order $\mu_1, \mu_2, \mu_3, \ldots, \mu_\delta$ was defined for the vertices of $SSD(P_n)$.

According to the arrangement of the vertices indicated below:

$$v^1_C \xrightarrow{\text{n-1}} v^1_{Rk} \xrightarrow{\text{n}} v^1_{L1} \xrightarrow{\text{n-1}} v^1_{R(k-1)} \xrightarrow{\text{n}} v^1_{L2} \xrightarrow{\text{n-1}} v^1_{R(k-2)} \xrightarrow{\text{n}} v^1_{L3}$$
$$\xrightarrow{\text{n-1}}, \ldots, \xrightarrow{\text{n}} v^1_{L(k-3)} \xrightarrow{\text{n-1}} v^1_{R3} \xrightarrow{\text{n}} v^1_{L(k-2)} \xrightarrow{\text{n-1}} v^1_{R2}$$
$$\xrightarrow{\text{n}} v^1_{L(k-1)} \xrightarrow{\text{n-1}} v^1_{R1} \xrightarrow{\text{n}} v^1_{Lk} \xrightarrow{\text{n}} v^2_{R1} \xrightarrow{\text{n-1}} v^2_{L(k-1)} \xrightarrow{\text{n+1}} v^2_{R(3)}$$
$$\xrightarrow{\text{n-1}} v^2_{L(k-3)} \xrightarrow{\text{n}} v^2_{R5} \xrightarrow{\text{n-1}} v^2_{L(k-5)}, \ldots, \xrightarrow{\text{n}} v^2_{R(k-5)} \xrightarrow{\text{n-1}} v^2_{L(5)}$$
$$\xrightarrow{\text{n}} v^2_{R(k-3)} \xrightarrow{\text{n-1}} v^2_{L(3)} \xrightarrow{\text{n}} v^2_{R(k-1)} \xrightarrow{\text{n-1}} v^2_{L(1)} \xrightarrow{\text{n}} v^3_{R(1)}$$
$$\xrightarrow{\text{n-1}} v^3_{L(k-1)} \xrightarrow{\text{n+1}} v^3_{R(3)} \xrightarrow{\text{n-1}} v^3_{L(k-3)} \xrightarrow{\text{n}} v^3_{R(5)}$$
$$\xrightarrow{\text{n-1}} v^3_{L(k-5)}, \ldots, \xrightarrow{\text{n}} v^3_{R(k-5)} \xrightarrow{\text{n-1}} v^3_{L(5)} \xrightarrow{\text{n}} v^3_{R(k-3)}$$
$$\xrightarrow{\text{n-1}} v^3_{L(3)} \xrightarrow{\text{n}} v^3_{R(k-1)} \xrightarrow{\text{n-1}} v^3_{L(1)}, \ldots, \xrightarrow{\text{n}} v^n_{R(1)} \xrightarrow{\text{n-1}} v^n_{L(k-1)}$$
$$\xrightarrow{\text{n+1}} v^n_{R(3)} \xrightarrow{\text{n-1}} v^n_{L(k-3)} \xrightarrow{\text{n}} v^n_{R(5)} \xrightarrow{\text{n-1}}, \ldots, \xrightarrow{\text{n}} v^n_{R(k-5)}$$
$$\xrightarrow{\text{n-1}} v^n_{L(5)} \xrightarrow{\text{n}} v^n_{R(k-3)} \xrightarrow{\text{n-1}} v^n_{L(3)} \xrightarrow{\text{n}} v^n_{R(k-1)} \xrightarrow{\text{n-1}} v^n_{L(1)}.$$

Therefore, the vertices of $SSD(P_n)$ can be assigned a label equivalent to the lower bound if the requirement of the theorem (3) is satisfied. This is true in both case (1) and case (2).
Hence, ϖ is a radio labeling.

Theorem 8: For $\alpha \in \mathbb{N}$, Then

$$rn(SSD(P_n)) = \begin{cases} n^2 - 3n - 2\alpha - \alpha q - \alpha n^2 + 2\alpha n \\ +2\alpha nq + 5 & \textit{if } n \textit{ is an even.} \\ n^2 - 3n - \alpha q - \alpha n^2 + 2\alpha nq - \alpha \\ +2\alpha n + 4 & \textit{if } n \textit{ is an odd.} \end{cases}$$

Proof: The proof is based on the Theorems (3), (5), (6), and (7) respectively.

Example 1: Figure (6) demonstrate both the arrangement (ordering) of the vertices as well as the optimal radio labeling of $SSD(P^5_8)$.

$$V^1_C \rightarrow V^1_{R7} \rightarrow V^1_{L1} \rightarrow V^1_{R6} \rightarrow V^1_{L2} \rightarrow V^2_{R6} \rightarrow V^2_{L2} \rightarrow V^3_{R6}$$
$$\rightarrow V^3_{L2} \rightarrow V^4_{R6} \rightarrow V^2_{L4} \rightarrow V^5_{R6} \rightarrow V^5_{L2} \rightarrow V^1_{R5} \rightarrow V^1_{L3}$$

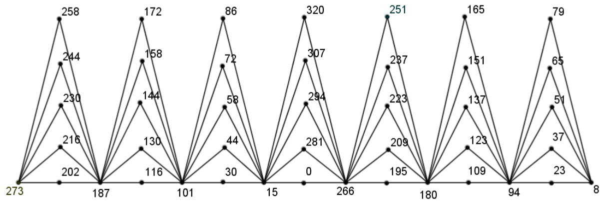

FIGURE 6. $SSD(P_8) = 320$ **where** $\alpha = 5$.

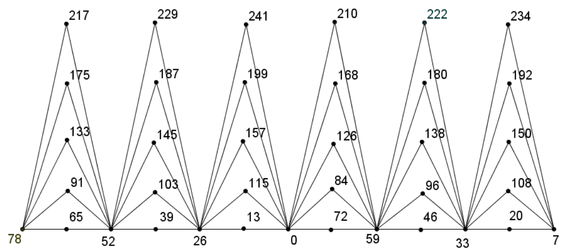

FIGURE 7. $SSD(P_7) = 241$ **where** $\alpha = 5$.

$$
\begin{aligned}
&\to V_{R4}^{1} \to V_{L4}^{1} \to V_{R4}^{2} \to V_{L4}^{2} \to V_{R4}^{3} \to V_{L4}^{3} \to V_{R4}^{4} \\
&\to V_{L4}^{4} \to V_{R4}^{5} \to V_{L4}^{5} \to V_{R3}^{1} \to V_{L5}^{1} \to V_{R2}^{1} \to V_{L6}^{1} \\
&\to V_{R2}^{2} \to V_{L6}^{2} \to V_{R2}^{3} \to V_{L6}^{3} \to V_{R2}^{4} \to V_{L6}^{4} \to V_{R2}^{5} \\
&\to V_{L6}^{5} \to V_{R1}^{1} \to V_{L7}^{1} \to V_{C}^{2} \to V_{C}^{3} \to V_{C}^{4} \to V_{C}^{5}.
\end{aligned}
$$

Example 2: Figure (7) demonstrate both the arrangement (ordering) of the vertices as well as the optimal radio labeling of $SSD(P_7^5)$.

$$
\begin{aligned}
V_C^1 &\to V_{R6}^{1} \to V_{L1}^{1} \to V_{R5}^{1} \to V_{L2}^{1} \to V_{R4}^{1} \to V_{L3}^{1} \to V_{R3}^{1} \\
&\to V_{L4}^{1} \to V_{R2}^{1} \to V_{L5}^{1} \to V_{R1}^{1} \to V_{L6}^{1} \to V_{R1}^{2} \to V_{L5}^{2} \\
&\to V_{R3}^{2} \to V_{L3}^{2} \to V_{R5}^{2} \to V_{L1}^{2} \to V_{R1}^{3} \to V_{L5}^{3} \to V_{R3}^{3} \\
&\to V_{L3}^{3} \to V_{R5}^{5} \to V_{L1}^{3} \to V_{R1}^{4} \to V_{L5}^{4} \to V_{R3}^{4} \to V_{L3}^{4} \\
&\to V_{R5}^{4} \to V_{L1}^{4} \to V_{R1}^{5} \to V_{L5}^{5} \to V_{R3}^{5} \to V_{L3}^{5} \\
&\to V_{R5}^{5} \to V_{L1}^{5}.
\end{aligned}
$$

VI. APPLICATION OF WIRELESS COMMUNICATION

See [10] for a detailed listing of all graph labelings and their applications. The use of radio labels in sensor networks enables fast communication. Radio labeling is used to assign a channel to each station to avoid interference. The smaller the distance between stations, the more interference occurs. To avoid interference, positive integers are assigned to the channels. The shorter distances between stations should result in stronger channel assignments. Each vertex in the radio label represents one transmitter. When two transmitters are adjacent, they are connected by an edge. Radio labeling is an efficient method for determining communication timing in sensor networks [49].

Industrial IoT Network for Manufacturing: In modern manufacturing plants, the Industrial Internet of Things (IIoT) plays a critical role in optimizing production processes. Super subdivision graphs can be used to model the factory layout, with machines, workstations, and sensors as vertices. Each machine and sensor is equipped with wireless communication capabilities.

You can use radio labeling techniques to assign unique labels to these devices based on their position in the graph. This labeling helps build an efficient wireless communication network within the factory. Machines can wirelessly communicate their status, maintenance needs and production data to a central control system. This real-time data exchange enables predictive maintenance, process optimization and reduced downtime.

By using supersub−division Graph's radio labeling in this industrial IoT application, manufacturers can increase productivity, reduce costs, and improve the overall efficiency of their operations.

VII. CONCLUSION

Wireless communication can sometimes be a very unpleasant experience when channel assignment problems occur when transmitting information over a distance using wireless technology. This annoyance is caused by simultaneous transmissions of a variety of information causing interference. It is possible to damage or disrupt communications when two good enough channels are available. One possible approach to reducing interference in a channel is to assign numbers to individual vertices within a graph. The edges connecting two vertices representing different channels indicate the links that exist between those channels. The use of radio labeling is necessary when different wireless media must be assigned to separate channels.

The goal of this paper is to provide insight into the use of radio labels in computer science, in particular their role in tasks such as channel assignment in wireless networks and network analysis using radio labels. This work addresses a number of mathematical situations, including situations where networks have geometric configurations, such as supersub−divisions of path graphs $SSD(P_n)$.

Radio labeling using supersub−division of path can be used for various applications in modern and future wireless networks, including the following: Network planning and optimization, 5G, IoT and smart cities, dynamic network reconfiguration, and emergency services. By using radio labeling based on supersub−division of paths, wireless networks can improve their performance, reliability, and functionality.

CONFLICT OF INTERESTS

The author(s) declare that there is no conflict of interests.

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